

Study Design Menu for Small Colleges

The Hackles Kit

Trust your hackles. Pause · Check · Tell.

FIELD COPY



PLATE I. The six strands, hand-inked, in the order the lessons take them.

Hackles · a free kit · quillsup.org

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Four designs a small college can run, with honest sample sizes. Every number here is a planning assumption computed by the code in `code/power_and_sample_size.py` (tests in `code/test_all.py`). None comes from a trial of Hackles, because no controlled trial of Hackles exists. Hackles has not been evaluated or approved by any agency. Nothing in this menu is legal advice; every study needs IRB review [IRB TO DETERMINE].

No student accounts, no student data collected, ever. In every design below, researchers receive aggregate teacher tallies, not student records.

01 **How to choose**

YOUR SITUATION	START WITH
One course section; you want to know if candidates gain confidence and skill	Design A (pre/post), with the honest label "no comparison group," or add a later cohort as comparison
Two or more K-12 classes taught by candidates or partner teachers; one group can wait a term	Design B (pre/post with a delayed-start comparison)
You can cover every class eventually, in steps, over two to four terms	Design C (stepped wedge)
You want to know why something works or fails; very few sites	Design D1 (qualitative case study)
You want to know whether teachers can deliver it as designed; 15 to 40 teachers	Design D2 (teacher-implementation study)

THE FIRST QUESTION IS always the one you can honestly answer. A small college cannot detect a small effect on students. It can learn a great deal about implementation, teacher confidence and what gets in the way.

What "small" means here. Course cohorts of 12 to 40 candidates, partner schools with 4 to 10 participating classes, 18 to 26 students per class.

Inputs that are assumptions, not evidence. Effect sizes (d, dz), the intraclass correlation (ICC) of class change scores, class size, and attrition. Replace them with your own values as soon as you have data, and report the range you used.

02 Sample size tables

Computed by `code/power_and_sample_size.py`, alpha .05 two-sided, power 80 percent unless stated. The simulation checks reproduce the targets (for example, 34 pairs for $d_z = 0.5$ gave simulated power of about 0.82).

A. One group, pre to post (paired t-test): pairs needed

DZ (MEAN CHANGE / SD OF CHANGES)	80% POWER	90% POWER
0.3	90	119
0.4	51	68
0.5	34	44
0.6	24	32
0.8	15	19
1.0	10	13

B. Two groups of individuals (for example, two cohorts of candidates): n per group, comparing change scores

D	80% POWER	90% POWER
0.3	176	235
0.5	64	86
0.8	26	34
1.0	17	22

C. Students in classes (two arms), analysis at the class level, m = 20 matched students per class

ICC OF THE CHANGE SCORE	D	DESIGN EFFECT	STUDENTS PER ARM	CLASSES PER ARM
0.02	0.3	1.38	243	13
0.02	0.5	1.38	89	5
0.02	0.8	1.38	36	2 (use 4)
0.05	0.3	1.95	344	18
0.05	0.5	1.95	125	7
0.05	0.8	1.95	51	3 (use 4)
0.10	0.3	2.90	511	26
0.10	0.5	2.90	186	10
0.10	0.8	2.90	76	4

Exact randomization tests need at least 4 classes per arm: with 3 per arm the smallest possible two-sided p is 0.100.

D. One yes/no item, matched pairs: pairs needed (McNemar)

SHARE WRONG TO RIGHT	SHARE RIGHT TO WRONG	NET GAIN	PAIRS
0.15	0.05	0.10	155
0.20	0.05	0.15	85
0.25	0.05	0.20	57
0.30	0.05	0.25	42
0.40	0.05	0.35	27

E. Class-level exact randomization test, simulated power (m = 20, ICC 0.05, 400 runs)

CLASSES PER ARM	D = 0.5	D = 0.8	D = 1.0
3	0.00	0.00	0.00
4	0.35	0.73	0.89
5	0.62	0.92	1.00
6	0.71	0.97	1.00

F. Stepped wedge, simulated power (m = 20, ICC 0.05, class-by-period variation SD 0.15 assumed, 200 runs)

CLASSES	STEPS	D = 0.3	D = 0.5	D = 0.8
6	3	0.24	0.53	0.91
9	3	0.37	0.76	0.99

What these tables say. With four to six classes per arm you can detect only large effects (about 0.8 standard deviations of student change). A stepped wedge with six classes has about 0.9 power for such an effect but only about half for a moderate one. These simulations assume no contamination between classes and no carry-over, so real power is likely lower. Plan to pool classes across terms or sites.

03 Design A: Pre/post (with or without a comparison group)

Question. Did participants' answers or skills change between the start and the end? With a later cohort as comparison: did they change more than a comparison group did over the same time?

Participants.

- Teacher candidates or in-service teachers in the college course, the 1-credit workshop or the PD (adults, with consent).
- Optional: K-12 classes taught by those teachers, reported only as aggregate class tallies (see Design B).

Measures. Academy Pre/Post Confidence Survey (R5; Part A mean, items 10 and 11 read separately; knowledge snapshot), observed skill (micro-teach rubric total; practicum checklist), course quiz mastery. All in the instruments file.

Procedure. Pre survey in Week 1; post survey in Week 13. If you add a comparison cohort (another section, a later term, or a waiting group), give them the same surveys at the same two times, with no course in between.

Sample size. Table A for one group. Table B for two cohorts. With a cohort of 24 candidates you can detect a change of about d_z 0.6 at 80 percent power; with 15 you can detect d_z 0.8. A cohort of 12 can detect only very large changes, so combine terms.

Strengths. Easy; fits the course; teachers get the information too. Confidence and knowledge are what the course targets.

Threats. No comparison group means maturation, other learning, repeated testing and wanting to please the instructor can all produce change. Ceiling effects if candidates start high. The instructor is also the person grading, which pushes toward socially desirable answers. Use a third party to collect and store the surveys, make participation voluntary and unrelated to grades, and release identifiable data only after grades are posted [IRB TO CONFIRM].

Honest claim. "Candidates' confidence scores rose by X on a 5-point scale ($n = 24$; no comparison group)."

04 Design B: Pre/post with a delayed-start comparison group (K-12 classes)

Question. Did students in classes that received the lessons first change more, on knowledge, confidence and Tell-Readiness, than students in similar classes that had not yet received them?

Participants. 4 to 8 classes at one or more partner schools (grade band chosen by the teachers), in two arms: taught first and delayed. Assign arms by coin flip or by the school's schedule; record how. The delayed classes receive the lessons after the second survey, so all students benefit.

Measures. The Assessment Pack student survey for the band (R2, R3 or R4), given by the teacher on paper with a teacher-held code, as ordinary

classroom practice. Researchers receive **class tallies only** (R7). Optionally the skill rubric as class counts (R6).

Procedure. Pre survey for all classes in Week 0; lessons in the taught-first classes over about ten weeks; post survey for all classes in Week 11; the delayed classes teach after. An optional delayed post for the taught-first classes at 4 to 8 weeks shows what lasted.

Sample size. Table C and E. With 20 matched students per class and ICC 0.05, 7 classes per arm give about 80 percent power for $d = 0.5$; 4 per arm give about 0.73 for $d = 0.8$. Four per arm is the minimum for an exact randomization test.

Analysis. Class-level difference in change, with randomization inference and a class bootstrap. See the analysis plan and the synthetic example (+24 points difference, 4 classes per arm).

Strengths. A comparison group controls for time and testing. Aggregate tallies protect students. Delayed start is fair.

Threats. Few classes; classes may differ at baseline; contamination between classes in the same grade; teacher effects; social desirability; the surveys have no reported validity evidence; PPRA, FERPA and consent rules apply [COUNSEL TO CONFIRM].

Honest claim. "The share of students giving the best answer on the core items rose more in classes that received the lessons first (+X points) than in delayed classes (+Y points); 8 classes; the interval is wide."

05 **Design C: Stepped wedge**

Question. Does the Tell-Readiness of students rise after their class begins the lessons, compared with the same classes before, and with other classes that have not yet begun?

Layout. Every class eventually receives the lessons. Classes cross over in steps (for example, 2 classes per step across 3 steps = 6 classes; 4 periods of measurement). Randomize the order of crossover if you can. Each class is measured at every period.

PERIOD	CLASS GROUP 1	CLASS GROUP 2	CLASS GROUP 3
1	control	control	control
2	lessons	control	control
3	lessons	lessons	control
4	lessons	lessons	lessons

Measures. The same class tallies (R7) at each period. Teachers give a short, identical survey each time, so a "pre" is the previous period's "post."

Sample size. Table F: 6 classes with 3 steps give about 0.9 power for $d = 0.8$ and about 0.5 for $d = 0.5$; 9 classes give about 0.76 for $d = 0.5$. Fewer than 6 classes makes randomization inference too coarse. Plan two to four terms.

Analysis. Treatment effect from a model with class and period effects (or the two-way fixed effects estimator on class-period means used in the simulation), with randomization inference over reassignments of classes to steps. A mixed model on student-level data needs data the privacy-by-design option does not provide.

Strengths. Every class gets the lessons; classes act as their own controls; fair; fits school schedules.

Threats. Time trends (students' answers may change across the year anyway); repeated testing; carry-over; staff turnover; the long timeline; contamination. The model assumes treatment effects are the same in every period.

06 Design D1: Qualitative case study

Question. How do candidates, instructors, mentor teachers and counselor partners experience the course, and what helps or gets in the way, especially in the high-care weeks and the field placement?

Sampling. One to three sites (course sections or partner schools), purposive: include a range of backgrounds, placements and comfort levels. Plan **8 to 12 interviews per role** and stop when new interviews add little. These are planning assumptions, not a rule [VERIFY with methods guidance for your approach].

Data.

- Semi-structured interviews (30 to 45 minutes) with candidates, the instructor, mentor teachers and the counselor partner.
- Documents: syllabus, course changes, the instructor's glow-and-grow notes, anonymous end-of-term feedback.
- Observation of class sessions, **excluding** the role-play rounds in Weeks 8 and 9 (observers would change the safe room). Observe the pre-briefing and the de-role and debrief only if all participants agree.
- No interview about any real child or incident. Ask about teacher practice and feelings in general terms.

Analysis. Thematic analysis by two coders; code a sample together; keep an audit trail; member checking with participants; report disconfirming cases.

Strengths. Finds problems that surveys miss; fits very small sites; no student data at all.

Threats. Few participants; instructor-researcher power; socially desirable answers; participants may reveal sensitive personal material (see the distress protocol); findings are not generalizable and should say so.

07 Design D2: Teacher-implementation study

Question. Can teachers deliver Hackles lessons as designed, with what adaptations, and with what support? Do they feel ready, and do they use the response routine correctly?

Participants. 15 to 40 teachers (adults, with consent) trained through the course, the workshop or the PD.

Measures.

- **Fidelity** to the ten non-negotiables in the Train-the-Trainer Guide, scored by a peer observer on the Fun, Safe, Skill-based rubric (Y, G or N per look-for).
- **Dosage:** lessons taught of ten, time per lesson, sessions skipped and why.
- **Adaptations** logged in a short teacher log (what changed, why).
- **Teacher confidence and readiness:** the Academy confidence survey (R5), response-readiness items 10 and 11, and the practicum checklist.
- **Aggregate counts** only for disclosures and referrals: the Teacher Feedback Form items (counts, no names; "check your policy before recording this on a form").
- **Barriers and supports** from short open responses.

Precision. Proportions are imprecise with small groups. Wilson 95 percent half-widths: $n = 10$, 26 points at 50 percent and 23 at 80 percent; $n = 20$, 20 and 17; $n = 30$, 17 and 14; $n = 40$, 15 and 12.

Strengths. Directly useful for improving training and for answering whether the model can be delivered. No student data.

Threats. Volunteer bias; observer effects; self-report; teachers who drop out are not observed; fidelity to a design is not evidence the design works.

Honest claim. "Of 24 teachers observed, 20 met at least 8 of 10 fidelity items; the most common adaptation was shortening the Discuss segment."

08 Common to all designs

- **Pre-register** the primary outcome and analysis (template provided). Label everything else exploratory.
- **IRB first.** Even aggregate-only designs need an IRB determination (exempt, not human subjects research, or full review) [IRB TO DETERMINE; COUNSEL TO CONFIRM].
- **Independence.** If the investigator wrote or teaches the course, add an independent person for recruitment, data custody and analysis checks. Declare conflicts, including any connection to the Hackles team.
- **Say plainly what Hackles is and is not.** It has not been evaluated in a controlled trial and has not been approved by any agency.
- **Report null and unwelcome results.** Disclosures that go up, or scores that go down, are information.
- **Protect small groups.** Do not report any subgroup under 10 students [VERIFY your district's suppression rule].
- **Share nothing identifiable.** No student names, codes or free text leave the school.
- **Plan for pooling.** Agree up front how classes across terms or sites will be combined, and keep the survey wording identical.

09 A worked planning example

A SMALL COLLEGE HAS 20 candidates per term and a partner school with 8 classes. The plan: Design A for candidates ($n = 20$ per term; combine two terms to reach about 40, enough for dz about 0.45), Design C for the school (6 to 8 classes, 3 to 4 steps, across two terms), and Design D1 with 10 interviews. The tallies are class-level only. The pre-registration says the primary outcomes are (1) candidates' Part A mean change and (2) students' Tell-Readiness change across class-periods. The limits statement says the study cannot detect small effects, cannot show behavior change, and cannot generalize beyond one school.

This menu is educational guidance, not legal advice.